

Unit 3: Data Collection Methods & Sampling Design Fundamentals

Handwritten Student Notebook

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1. Methods of Data Collection

Source: C.R. Kothari, Research Methodology (2e), Ch. 6, pp. 95–112

The gathering of empirical data is central to any scientific study. Based on origin and acquisition method, statistical data is categorized into **Primary Data** and **Secondary Data**.

Definition:

Primary Data vs. Secondary Data

- **Primary Data:** Data collected afresh and for the first time by the researcher directly from primary sources. It possesses an original character (e.g., experimental sensor logs, direct user interview recordings, field observations).
- **Secondary Data:** Data that has already been collected, compiled, and analyzed by someone else for another primary purpose (e.g., government census reports, RBI annual bulletins, published benchmark datasets).

1.1 Primary Data Collection Methods

In descriptive, behavioral, and experimental surveys, primary data is collected through four dominant modalities:

Method	Operational Mechanism	Merits	Demerits & Limitations
1. Observation Method	Systematic viewing, recording, and analysis of subjects' behavior without asking questions. Can be structured/unstructured, participant/non-participant.	• Eliminates respondent recall bias. • Independent of respondent's willingness to report. • Captures current natural behavior.	• Expensive and time-consuming. • Cannot reveal past history, internal motives, or unexpressed feelings. • Prone to observer subjectivity.
2. Interview Method	Direct verbal interaction and question-and-answer exchange. Can be Personal Interviews (structured/unstructured) or Telephonic Interviews.	• Allows in-depth probing and complex follow-up questions. • Non-verbal reactions can be observed. • Very high response rate compared to mail surveys.	• Costly; requires trained, skilled interviewers. • Vulnerable to interviewer bias and social desirability distortion. • Slower execution over large geographic areas.

Method	Operational Mechanism	Merits	Demerits & Limitations
3. Questionnaires	Structured set of written questions mailed, emailed, or administered online to respondents for independent completion.	• Low cost even over vast geographical coverage. • Free from interviewer bias. • Respondents have ample time to give well-thought-out answers.	• Notoriously low response rate (often < 20–30%). • Usable only when respondents are literate. • Inflexible; ambiguous questions cannot be clarified.
4. Schedules	Standardized proforma of questions carried and filled in personally by trained enumerators/field investigators.	• Can be administered to illiterate populations. • Almost 100% completion rate. • Identity of respondent is verified on the spot.	• Highly expensive and logistically intensive. • Requires rigorous training to avoid enumerator bias.

1.2 Guidelines for Constructing an Effective Questionnaire

A poorly drafted questionnaire invalidates survey results before analysis begins. Kothari prescribes five Golden Rules of questionnaire design:

1. **Brevity & Focus:** Keep questions strictly minimal and direct. Every question must tie to a specific research hypothesis.
2. **Logical Psychological Sequencing:** Start with simple, engaging introductory questions; place sensitive, controversial, or personal demographic questions at the end.
3. **Neutral, Unambiguous Phrasing:** Avoid double negatives, emotionally loaded terms, and technical jargon. (e.g., avoid "Do you not agree that...").

4. **Appropriate Question Format:** Balance structured closed-ended items (Multiple Choice, Likert Rating) for quantitative statistical aggregation with selective open-ended items for qualitative insights.
5. **Mandatory Pre-testing (Pilot Study):** Always test the draft questionnaire on a small pilot sample (15–20 subjects) to identify confusing wording, ambiguous choices, and completion latency.

1.3 Collection of Secondary Data & Mandatory Precautions

Secondary data saves enormous financial and temporal resources, but a researcher must never accept it uncritically. Kothari specifies three mandatory verification criteria:

- **Reliability:** Who compiled the data? Was the agency reputable, competent, and free from commercial bias? Were the measuring tools scientifically sound?
- **Suitability:** Does the secondary data match the operational definitions, scope, and objective of the current investigation?
- **Adequacy:** Is the dataset sufficiently broad and temporally current? If circumstances have shifted fundamentally (e.g., pre-cloud vs. cloud-native telemetry), older datasets are inadequate.

2. Sampling Fundamentals & Steps in Sample Design

Source: C.R. Kothari, Research Methodology (2e), Ch. 4, pp. 55–58

In any empirical enquiry, the researcher must decide whether to examine every member of the target population or a scientifically selected subset.

Attribute	Census Method (Complete Enumeration)	Sample Survey Method
Coverage	Every single individual unit of the entire universe is inspected.	Only a representative subset (sample) is inspected.
Resource Demand	Enormous financial budget, huge workforce, very long timeline.	Fractional cost, minimal manpower, rapid turnaround.
Feasibility	Impossible when universe is infinite, accessible units are restricted, or tests are destructive.	Universal feasibility; the only method possible for destructive testing.

Attribute	Census Method (Complete Enumeration)	Sample Survey Method
Accuracy & Error	Zero sampling error; however, very high non-sampling errors (enumerator fatigue, recording errors).	Contains measurable sampling error (SE), but minimizes non-sampling error due to superior supervision.

2.1 The Seven Steps in Developing a Sample Design

A **sample design** is a definite plan determined before any data is collected for obtaining a sample from a given population. Kothari defines seven sequential steps:

Step 1: Type of Universe

Define the universe of study. It can be Finite (fixed countable number of items, e.g., students enrolled in MSRIT) or Infinite (unbounded stream of items, e.g., network packets arriving at a router, stars in a galaxy).

Step 2: Sampling Unit

The elementary unit chosen for sampling. It can be geographical (state, district, village), structural (school, hospital, data center), or individual (student, patient, CPU core).

Step 3: Source List (Sampling Frame)

The physical list containing the names and identifiers of all units in the universe from which the sample is drawn. It must be exhaustive, strictly up-to-date, accurate, and completely free from omission or duplication.

Step 4: Size of Sample (n)

The number of items to be selected from the universe. The sample size must be optimum: large enough to ensure desired statistical precision and power, but within budgetary constraints.

Step 5: Parameters of Interest

Deciding which population parameters are to be estimated (e.g., population mean μ , proportion P , or variance σ^2). The expected variance directly impacts the required sample size.

Step 6: Budgetary Constraints

Cost considerations exercise a major control over sample size and methodology. The design must deliver maximal precision per rupee spent.

Step 7: Sampling Procedure

The explicit probability technique or algorithm by which the sampling units are selected (Simple Random Sampling, Stratified Sampling, Systematic Sampling, or Cluster Sampling).

3. Criteria & Characteristics of a Good Sample Design

Source: C.R. Kothari, Research Methodology (2e), Ch. 4, pp. 58-60

3.1 Criteria for Selecting a Sampling Procedure

When selecting a sampling procedure, the researcher must identify and minimize two fundamental types of errors: **Sampling Error** and **Systematic Bias**.

Definition:

Systematic Bias (Non-Sampling Error)

Systematic bias results from procedural errors in the design that consistently skew the sample estimates in one direction. Unlike sampling error, systematic bias does not decrease as sample size increases. Primary causes include:

1. **Inappropriate Sampling Frame:** If the source list omits an entire segment of the target universe (e.g., using a landline directory to sample modern smartphone users), the frame is structurally biased.
2. **Defective Measuring Device:** Biased survey question wording, poorly calibrated laboratory instruments, or leading prompts.
3. **Non-Respondents:** Occurs when individuals who refuse or fail to respond differ significantly in characteristics from those who participate.
4. **Indeterminacy Principle:** The very act of observing or questioning an individual alters their natural behavior or stated opinions.
5. **Natural Bias in Reporting:** Systematic respondent tendency to over-report socially approved behaviors (e.g., donations, income, study hours) and under-report negative behaviors.

3.2 Characteristics of a Good Sample Design

To be considered scientifically sound, a sample design must possess the following five indispensable characteristics:

1. **True Representativeness:** The sample must accurately mirror the distribution, proportions, and diversity of the parent population across all relevant parameters.
2. **Controlled & Small Sampling Error:** Must produce sample statistics that yield narrow confidence intervals and high statistical precision for given costs.

3. **Systematic Bias Minimization:** The design must incorporate procedural safeguards (e.g., randomization, validated frames, follow-up on non-respondents) to eliminate structural bias.
4. **Economic & Operational Feasibility:** Must be realistic to execute within available financial budgets, computational capacity, and scheduled calendar deadlines.
5. **Statistical Generalizability:** The sample findings must be safely and reliably applicable to the universe with mathematically quantifiable confidence levels ($1 - \alpha$).

4. Quick Revision Sheet

CIE-1 Rapid Memory Anchors

Concept	Quick Definition / Key Formula	Exam Catchphrase
Primary Data	First-hand data gathered afresh by researcher.	Original in character, direct evidence.
Secondary Data	Previously compiled data by another agency.	Must verify: Reliability, Suitability, Adequacy.
Observation Method	Systematic viewing and recording of behavior without asking questions.	No recall bias; observer bias present.
Interview Method	Verbal interaction and probing exchange.	High response rate; expensive; interviewer bias.
Questionnaire vs Schedule	Questionnaire mailed to literate respondents; Schedule carried by trained enumerator.	Schedule has higher completion rate and verifies identity.
Census vs Sample	Census: 100% complete enumeration. Sample: Representative subset.	Sampling mandatory for infinite universe and destructive testing.
Sampling Frame	Source list containing all sampling units in the universe.	Must be exhaustive, current, and duplication-free.
Systematic Bias	Structural design error skewing estimates in one direction.	Does NOT diminish with increasing sample size n .

Concept	Quick Definition / Key Formula	Exam Catchphrase
Good Sample Design	Representative, small sampling error, viable cost, generalizable.	Five gold criteria of Kothari Ch. 4.

5. Authentic Exam Questions & Solutions

Source: CIE-2 2025, SEE 2024-2026, Makeup Papers

Q1. Compare and contrast the observation method and interview method of primary data collection. (8 Marks · CIE-2 2025 Q1a, SEE 2024 Q5a)

Answer: Observation systematically records current behavior without verbal cooperation, eliminating respondent recall bias, but cannot capture internal motives and is prone to observer bias. The interview method involves direct verbal exchange, allowing in-depth probing into personal attitudes, but suffers from respondent recall bias and interviewer bias.

Q2. Explain the seven steps involved in developing a sample design. (8 Marks · CIE-2 2025 Q3b, SEE Jan 2026 Q6a)

Answer: (1) Type of universe (finite vs infinite), (2) Sampling unit, (3) Source list / sampling frame, (4) Size of sample (n), (5) Parameters of interest, (6) Budgetary constraints, and (7) Sampling procedure.

Q3. What are the characteristics of a good sample design? (8 Marks · SEE Jan 2026 Q6a)

Answer: True representativeness, controlled and small sampling error, minimization of systematic bias, economic and operational feasibility, and statistical generalizability with calculable confidence levels.

Q4. What precautions must a researcher take before using secondary data? (8 Marks · Makeup 2024 Q5a)

Answer: Must verify (1) **Reliability** (integrity and competence of publishing agency), (2) **Suitability** (match with current operational definitions and scope), and (3) **Adequacy** (geographic breadth and temporal relevance).

6. Prescribed References

- **C.R. Kothari & Gaurav Garg:** Research Methodology: Methods and Techniques, 4th/2nd Edition, New Age International Publishers, 2019. Chapters 4 and 6.
- **David V. Thiel:** Research Methods for Engineers, Cambridge University Press, 2014. Chapters 1, 2, and 3.
- **Faculty Course Material:** Course Code AL58, Department of Information Science & Engineering, Ramaiah Institute of Technology.